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A MAJOR PROJECT PROPOSAL ON

AUTOMATIC FACIAL RECOGNITION BASED ATTENDANCE SYSTEM

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Abstract

This project proposes an Automatic Facial Recognition Based Attendance System (AFRAS) a modern, intelligent, and contactless solution to the operational challenges of traditional manual roll calls in educational institutions. Manual methods waste valuable teaching time and are prone to errors and manipulation. AFRAS aims to develop a fully automated attendance system using facial recognition with an intuitive web-based dashboard for management. The system leverages computer vision and machine learning (Python, OpenCV, Dlib), employing deep learning models for face encoding and recognition. The methodology is Agile + Incremental. Expected outcomes include significant time reduction, elimination of proxy attendance, and provision of secure, centralized digital records for administrators, teachers, and students.

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1. INTRODUCTION

An Automatic Facial Recognition Based Attendance System (AFRAS) is a modern, intelligent alternative to traditional attendance system by leveraging computer vision and machine learning techniques to identify individuals in real time. Instead of relying on physical tokens or manual verification, AFRAS uses a camera feed to detect faces, extract facial features, and match them against a pre-registered database, and automatically record attendance with timestamps and identity markers. This minimizes human involvement, eliminates common loopholes, and drastically improves both efficiency and reliability. This project will implement AFRAS in Python, focusing on reliability, privacy-aware design, and ease of deployment for classroom or small-office environments.

2. PROBLEM STATEMENT

In many educational institutions, the process of taking student attendance remains a significant operational challenge. Traditional manual roll calls consume valuable teaching time (5-10 minutes per lecture) and are prone to errors and deliberate manipulation. While digital systems like RFID cards or fingerprint scanners reduce some manual work, they are not foolproof; cards can be shared, and fingerprint scanners require physical contact, leading to hygiene concerns and wear-and-tear. The lack of an automated, secure, and contactless system makes it difficult to prevent proxy attendance and ensure accurate, real-time record-keeping. There is a pressing need for an intelligent system that can accurately verify student identity, automate the logging process, and integrate seamlessly with existing management systems, thereby saving time, reducing administrative burden, and enhancing security.

3. OBJECTIVES

The main objective of this project is to develop a fully automated attendance system using facial recognition along with an intuitive web-based dashboard for system management.

Scope of Project:

The Face Recognition Attendance System is a web-based tool designed to enhance the operational efficiency of educational institutions. For Administrators, it provides tools to manage student profiles, courses, and generate comprehensive attendance reports. Teachers can initiate and stop attendance sessions for their respective classes and view real-time attendance logs. Students are automatically marked present when recognized by the system and can view their personal attendance history. The core features include student registration via facial data capture, real-time face detection and recognition, a secure central database, and a role-based web dashboard. The system is designed with scalability and security in mind, ensuring data privacy through encryption. Its primary aim is to promote a modern, efficient, and contactless academic environment

4. METHODOLOGY

4.1 System Development Methodology

The Agile + Incremental methodology will be used for this system. This approach combines the flexibility of agile development with the structured delivery of the Incremental model. The project will be divided into small, functional increments such as Face Detection, Database Integration, and Web Dashboard each delivering a working piece of software. Agile principles, such as iterative development, continuous feedback, and collaboration, will be applied within each increment. This allows for early testing of core functionalities like the recognition algorithm, adaptability to changing requirements, and continuous improvement. Stakeholders can see demonstrable progress early on, while additional features are added incrementally. This method is highly suitable for an AI-based project with complex components, as it ensures timely delivery, reduces risks, and aligns the final product with user needs.

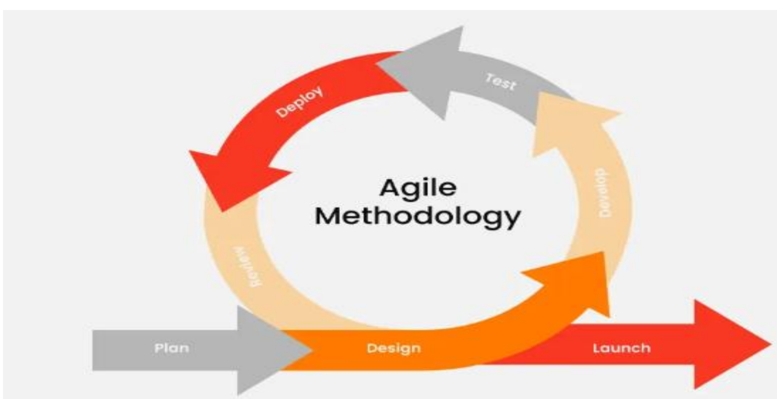

Fig.1: Agile method for the project development

4.2 Requirement Identification

4.2.1 Literature Review

The research article, "Face Recognition based Attendance System," reviews various techniques for automating student attendance. Face recognition is preferred over other biometrics like fingerprint scanning due to its non-contact process. Early challenges included performance in uncontrolled conditions and preventing cheating. To address fraud, one system used a dual vision camera and 3WPCA, achieving 98% anti-cheating accuracy. Machine learning algorithms discussed include SVM, MLP, and CNN, with CNN achieving the highest test accuracy of 98% on a self-generated database. Processing steps involve pre-processing (like histogram equalization) , feature extraction using PCA, LDA, or LBP, and matching features with a database. The resulting system automates attendance and generates reports. [1]

The research article, "An Automated Attendance System Using Facial Detection and Recognition Technology," proposes a solution to the inefficiencies, human errors, and proxy attendance issues inherent in manual classroom attendance recording. The system leverages Facial Detection and Recognition Technology for a robust and automated process. The core methodology employs Python for programming, OpenCV for image processing tasks, and the Local Binary Pattern Histogram (LBPH) algorithm for feature extraction and face recognition, citing its strength in varied lighting conditions. Haar Cascade classifiers are utilized for rapid face detection within captured frames. Implemented using low-cost hardware like the Raspberry Pi, the system captures live images, compares extracted features against a centralized database, and automatically updates the attendance record in real time. The paper concludes that this automated, cost-effective, and efficient system offers a reliable replacement for conventional methods, significantly streamlining the attendance process and improving data accuracy.[2]

The research article, "Automatic Face Recognition System Using Deep Convolutional Mixer Architecture and AdaBoost Classifier," addresses the need for robust and accurate face recognition, especially in environments with variations in pose, illumination, and expression. The authors propose a novel hybrid model, the Deep Convolutional Mixer Architecture (DCMA), which integrates the feature extraction strengths of Convolutional Neural Networks (CNNs) with the token-mixing capabilities of MLP-Mixers. The DCMA effectively extracts deep, distinctive features from face images. These extracted features are then fed into an AdaBoost classifier for final recognition, which helps to minimize misclassification errors. The proposed DCMA-AdaBoost system was evaluated against benchmark datasets like LFW, YouTube Faces (YTF), and Extended Yale-B, demonstrating superior performance. The system achieved a classification accuracy of 99.50% on LFW, confirming the effectiveness of combining deep learning for feature extraction with a strong ensemble classifier for decision-making.[3]

4.2.2 Requirement Analysis

Functional Requirements:

1. Face Registration Module: Capture multiple facial images of a student, extract encodings, and store them in the database linked to their profile.

2. Real-Time Recognition Module: Access webcam feed, detect faces, compare them with the stored encodings, and identify the student.

3. Attendance Logging Module: Automatically mark attendance in the database with a student ID, timestamp, and subject/class upon successful recognition.

4. Admin Module:

User Management – Create/edit student and teacher accounts.

Course & Subject Management – Link attendance sessions to specific courses.

Reports – Generate and export attendance reports (daily, weekly and monthly).

5. Teacher Module:

View Attendance – See real-time logs and reports for their classes.

6. Security & Access Control:

Role-Based Access – Restrict functionalities based on user roles (Admin, Teacher, Student).

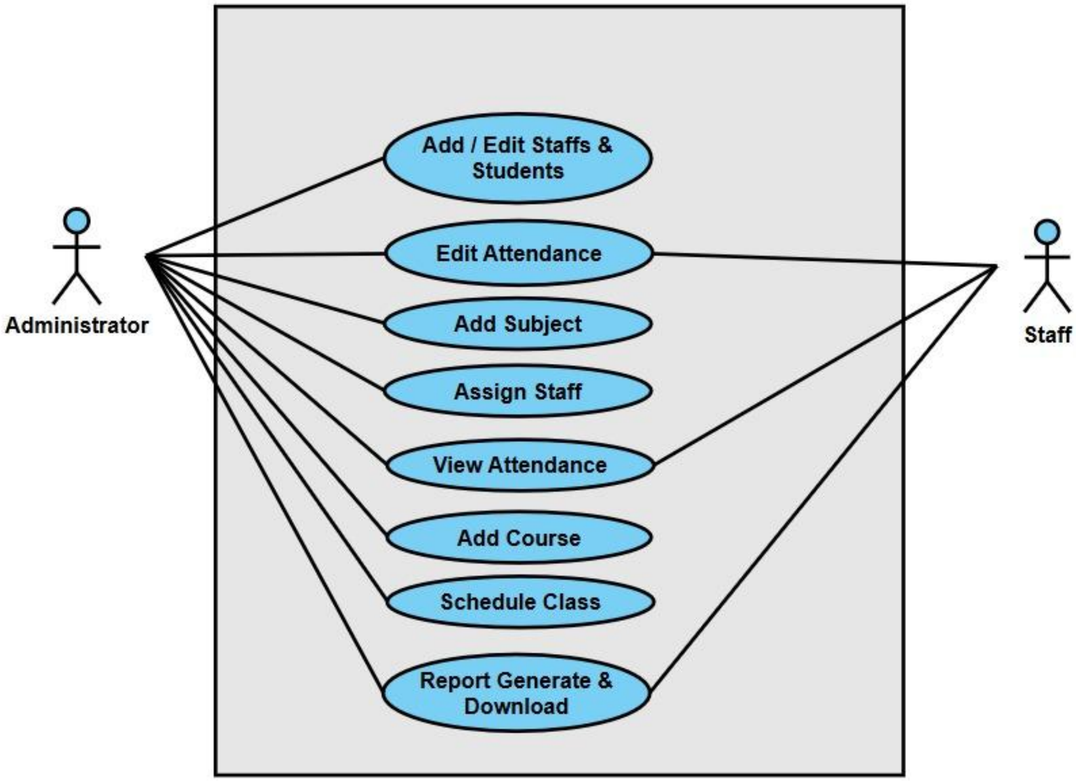

Fig.2: Use case diagram of AFRAS

Non-Functional Requirements:

1. Performance Requirements: The system must process video frames in real-time (≥15 FPS) and have a response time of less than 3 seconds for recognition and logging. It should support multiple concurrent recognition sessions.

2. Security Requirements: All facial encodings and personal data must be encrypted. The system must ensure users can access only their authorized information.

3. Accuracy Requirements: The system should achieve a recognition accuracy of over 90% under normal lighting conditions with a low False Acceptance Rate (FAR).

4. Scalability Requirements: The database design should support thousands of student records and allow for cloud hosting to facilitate future expansion.

4.3 Feasibility Study

4.3.1 Technical Feasibility

The project is technically feasible. The core technologies Python, OpenCV, Dlib, and Django are mature, well-documented, and open-source. These libraries provide robust, pre-trained models for face detection and recognition, significantly reducing development complexity. Hardware requirements (a standard webcam and a modern computer) are commonly available. The chosen tech stack is capable of supporting the system's real-time processing and data management requirements.

4.3.2 Operational Feasibility

The system is designed to significantly simplify the attendance process for all stakeholders. Its automated nature will save time for teachers and reduce administrative burden. The web-based dashboard offers a user-friendly interface. By providing immediate, transparent access to records, it aligns perfectly with the goals of improving operational efficiency and accountability in educational institutions, making it highly operationally feasible

4.3.3 Economic Feasibility

The system is economically feasible. Development will utilize open-source software and libraries, eliminating licensing costs. The hardware requirement (a webcam) is a one-time, low-cost investment. The long-term benefits, including saved man-hours, reduced paperwork, and elimination of proxy attendance, provide a strong return on investment for institutions, outweighing the initial development and setup costs. We are expecting economic support from our department for following hardware requirements needed in our project:

SN	Item	Estimated Cost (Rs)
1	Web-Cam	3000
2	Connecting Cables	200
Total		**3200**

4.3.4 Schedule

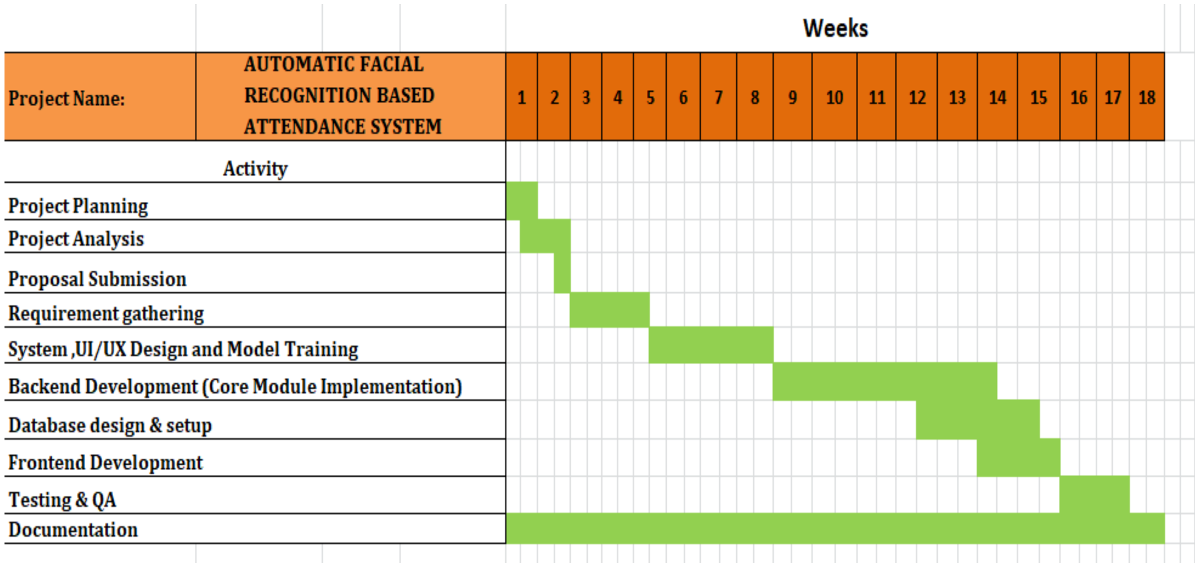

Fig.3: Gantt chart of time schedule

4.4 High Level Design of System

4.4.1 System flow diagram

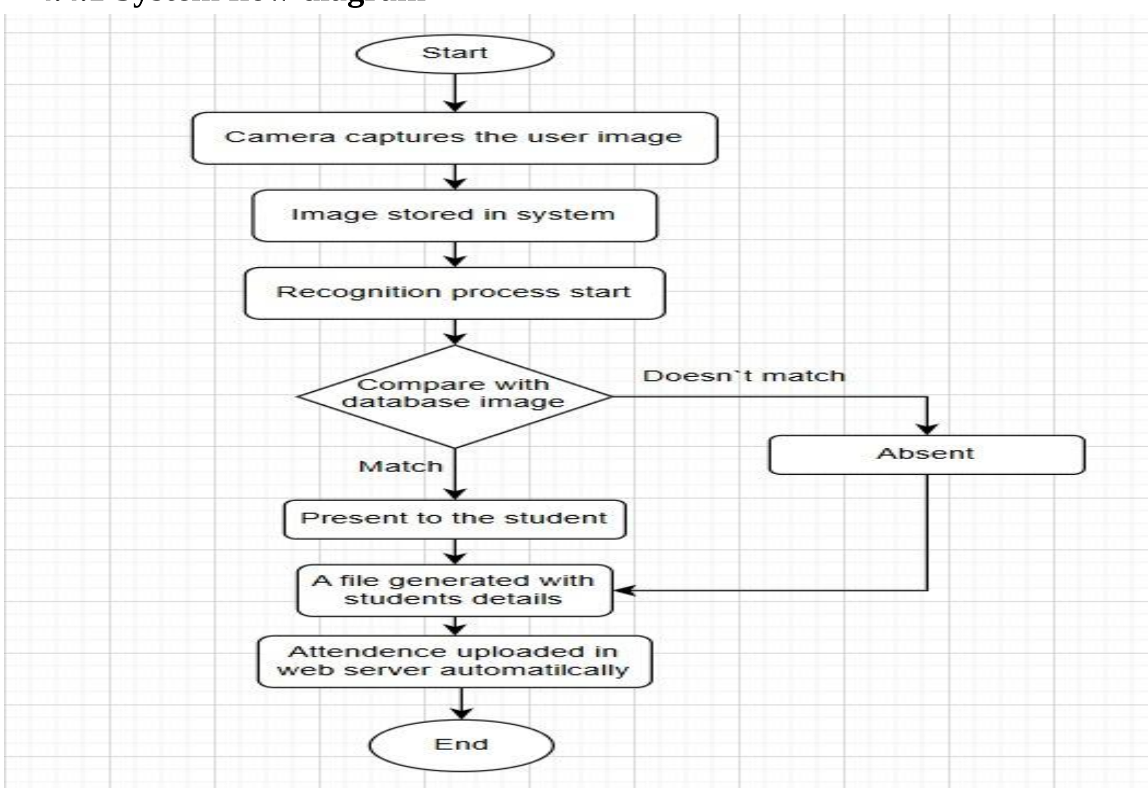

Fig.4: System Flow Diagram of proposed system

4.4.2 Algorithm Description

1. Face Detection (Haar Cascades or CNN-based Detectors):

This is the first step, where the algorithm scans the input image from the webcam to identify regions that contain a human face. Haar Cascade, proposed by Viola and Jones, is a machine learning-based approach that uses a cascade of simple features to rapidly detect faces in real-time. Alternatively, more accurate Convolutional Neural Network (CNN**)** based detectors like those in Dlib can be used. This step outputs the coordinates of the bounding box around each detected face.

2. Face Encoding and Recognition (Deep Learning Models):

For each detected face, the system generates a numerical representation, or "face encoding." This is typically done using a pre-trained Deep Learning model (e.g., Dlib's ResNet-based model or FaceNet). The model converts the facial image into a 128-dimensional (or similar) vector that uniquely represents the facial features. Recognition is performed by comparing the encoding of the detected face with the encodings of all registered students stored in the database using a distance metric (e.g., Euclidean distance). The registered face with the smallest distance below a pre-defined threshold is considered a match.

5. EXPECTED OUTCOME

Upon full development and deployment, the system will significantly reduce time spent on daily attendance tasks while eliminating manual errors and proxy attempts. Its contactless design ensures a hygienic, efficient check-in process, and the centralized digital records offer transparency and easy access for students, faculty, and administrators. Beyond streamlining operations, it establishes a strong foundation for broader institutional automation and future AI-driven enhancements in academic management.

6. REFERENCE

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